UNIT 9

Amines

Commercially, amines matter most as intermediates on the way to medicines and fibres.

By the end of this unit

- picture amines as ammonia derivatives with a pyramidal shape
- sort amines into primary, secondary and tertiary
- give amines their common and IUPAC names
- outline the main routes for making amines
- explain how amines behave
- tell primary, secondary and tertiary amines apart
- explain how diazonium salts are made, and why they matter for building aromatic compounds, azo dyes among them

Where amines turn up

- **In nature** — within proteins, vitamins, alkaloids and hormones
- **Made by industry** — polymers, dyestuffs and drugs
- **Adrenaline**, **ephedrine** — biologically active, each with a secondary amino group; given to raise blood pressure
- **Novocain** — synthetic amino compound; dental anaesthetic
- **Benadryl** — familiar antihistamine; carries a tertiary amino group
- **Quaternary ammonium salts** — work as surfactants
- **Diazonium salts** — intermediates for many aromatic compounds, dyes included

The unit runs in two parts: **I · Amines**, then **II · Diazonium salts**.

I · Amines

What counts as an amine

- Start from NH_3 and put alkyl and/or aryl groups in place of one, two or all three of its hydrogen atoms — each result is an amine.

$CH_3–NH_2$ $C_6H_5–NH_2$ $CH_3–NH–CH_3$ $CH_3–N(CH_3)_2$

A.1 Structure of amines

- Just as in ammonia, the nitrogen is **trivalent** and holds one **unshared electron pair**.
- Its orbitals are therefore ***sp*^{3} hybridised**, and the amine is **pyramidal**.
- Three sp^3 orbitals overlap with orbitals of H or C, according to what the amine is built from; in every amine the fourth carries the unshared pair.
- That unshared pair affects the C–N–E bond angle — continued on sheet 2.

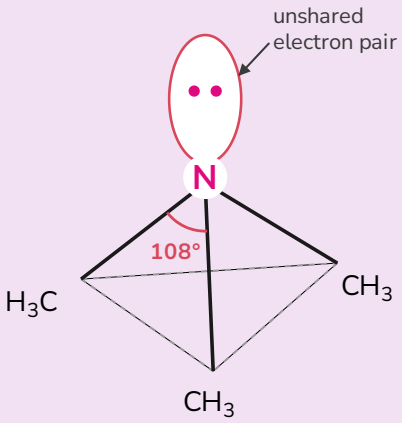


Figure 9.A Pyramidal shape of trimethylamine

The angle at nitrogen

- with E standing for C or H, the C–N–E angle falls short of the tetrahedral 109.5°
- in trimethylamine it is 108° (Figure 9.A)
- the unshared pair sits in the fourth sp^3 orbital, at the apex above the three CH_3 groups

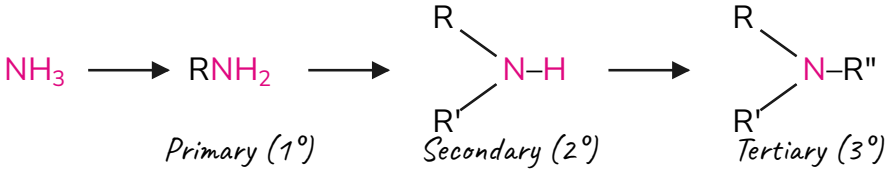
A.2 Classification

Counting the replaced H atoms

- primary (1°), secondary (2°) or tertiary (3°) — set by how many H atoms of NH_3 an alkyl or aryl group has taken over
- one H swapped for R or Ar gives RNH_2 or $ArNH_2$, a **primary amine**
- two H of NH_3 replaced — or one H of R–NH_2 replaced by a further group R' — gives R–NHR', a **secondary amine**; R' may match R or differ from it
- one more H replaced by an alkyl/aryl group gives a **tertiary amine**

Simple or mixed

- **simple** — every alkyl or aryl group on N is the same
- **mixed** — the groups are not all the same



A.3 Nomenclature

Common system

- aliphatic amine: alkyl group name placed before "amine", written as one word — **alkylamine**, e.g. methylamine
- secondary/tertiary amines with two or more identical groups: **di** or **tri** goes in front of the alkyl name

IUPAC — primary amines

- named as **alkanamines**
- the final 'e' of the alkane name gives way to "amine": CH_3NH_2 → methanamine

More than one –NH_2 group

- number the chain carbons carrying –NH_2 to fix their positions
- put di, tri, … before "amine"; here the hydrocarbon keeps its final 'e'
- H_2N–CH_2–CH_2–NH_2 → ethane-1,2-diamine

Secondary & tertiary: the N locant

- the locant **N** marks any substituent attached to the nitrogen atom
- worked case: $CH_3NHCH_2CH_3$ — its name follows on sheet 3

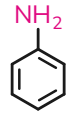
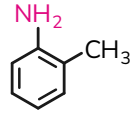
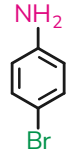
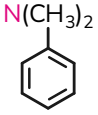
The N locant at work

- $CH_3NHCH_2CH_3$ → **N-methylethanamine**
- $(CH_3CH_2)_3N$ → **N,N-diethylethanamine**
- further cases are collected in Table 9.A

Arylamines

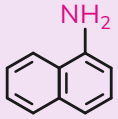
- the $-NH_2$ group is bonded straight to a benzene ring; $C_6H_5NH_2$ is the simplest one
- common name **aniline** — a name IUPAC also accepts
- IUPAC route: the arene's final 'e' is replaced by 'amine', so $C_6H_5-NH_2$ is **benzenamine**

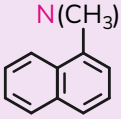
TABLE 9.A **Nomenclature of Some Alkylamines and Arylamines**

Amine	Common name	IUPAC name
$CH_3-CH_2-NH_2$	Ethylamine	Ethanamine
$CH_3-CH_2-CH_2-NH_2$	*n*-Propylamine	Propan-1-amine
$CH_3-\underset{\displaystyle NH_2}{\underset{\vert}{CH}}-CH_3$	Isopropylamine	Propan-2-amine
$CH_3-\underset{\displaystyle H}{\underset{\vert}{N}}-CH_2-CH_3$	Ethylmethylamine	N-Methylethanamine
$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{N}}-CH_3$	Trimethylamine	N,N-Dimethylmethanamine
$C_2H_5-\overset{1}{N}-\overset{2}{CH_2}-\overset{3}{CH_2}-\overset{4}{CH_2}-CH_3$ (with C_2H_5 on N)	*N,N*-Diethylbutylamine	N,N-Diethylbutan-1-amine
$NH_2-\overset{1}{CH_2}-\overset{2}{CH}=\overset{3}{CH_2}$	Allylamine	Prop-2-en-1-amine
$NH_2-(CH_2)_6-NH_2$	Hexamethylenediamine	Hexane-1,6-diamine
	Aniline	Aniline or Benzenamine
	*o*-Toluidine	2-Methylaniline
	*p*-Bromoaniline	4-Bromobenzenamine or 4-Bromoaniline
	*N,N*-Dimethylaniline	N,N-Dimethylbenzenamine

In-text questions

T1 Say whether each amine below is primary, secondary or tertiary.

(i) 

(ii) 

(iii) $(C_2H_5)_2CHNH_2$

(iv) $(C_2H_5)_2NH$

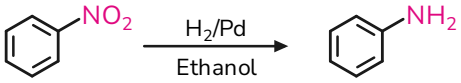
T2 (i) Draw every isomeric amine that fits the formula $C_4H_{11}N$.
(ii) Give the IUPAC name of each isomer.
(iii) Name the type of isomerism that relates the different pairs.

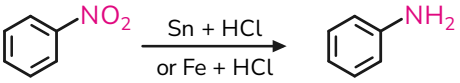
A.4 Preparation of amines

The routes used to make amines are laid out below.

1 · Reducing nitro compounds

- nitro compounds become amines with H_2 gas over finely divided **Ni, Pd or Pt**, or with a **metal in acid**
- nitroalkanes are reduced in the same way to the matching alkanamines
- **iron scrap + HCl** is the favoured pair: the $FeCl_2$ produced hydrolyses and releases HCl as the reaction goes on, so only a little HCl is needed to start it

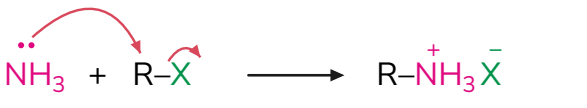
(i) 

(ii) 

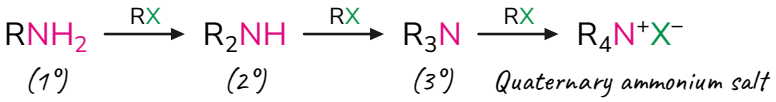
2 · Ammonolysis of alkyl halides

- a nucleophile readily breaks the C–X bond of alkyl or benzyl halides (see Unit 6, Class XII)
- with **ethanolic NH_3** such a halide undergoes nucleophilic substitution — the halogen is replaced by $–NH_2$
- breaking the C–X bond with an NH_3 molecule is called **ammonolysis**; it is run in a **sealed tube at 373 K**
- the primary amine formed is itself a nucleophile, so it attacks more alkyl halide — giving 2° and 3° amines and, last of all, a **quaternary ammonium salt**

Ammonia as the attacking nucleophile



Ammonolysis keeps going



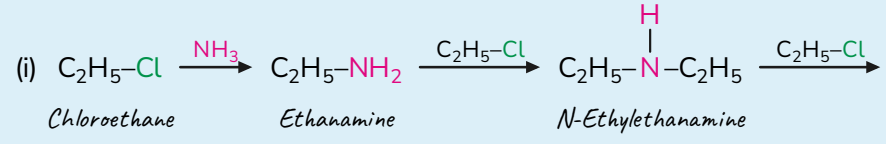
Freeing the amine · the drawback

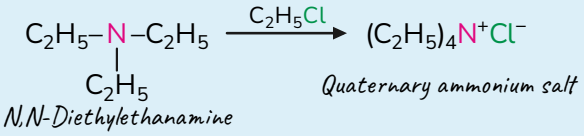
- a strong base releases the free amine from its ammonium salt (equation below)
- drawback: the product is a **mixture** of 1°, 2° and 3° amines plus a quaternary ammonium salt
- a **large excess of NH_3** makes the primary amine the main product
- halide reactivity towards amines: **RI > RBr > RCl**

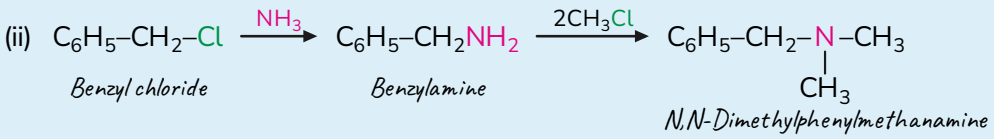


Worked example 9.a

Give the equations for (i) ethanolic NH_3 acting on C_2H_5Cl; (ii) ammonolysis of benzyl chloride, followed by reaction of the amine formed with two moles of CH_3Cl.

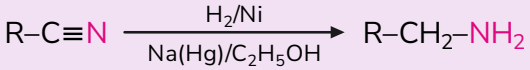






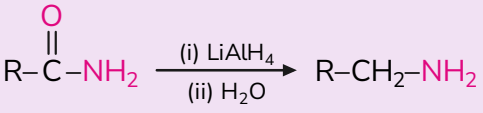
3 · Reducing nitriles

- reduction with **$LiAlH_4$** or by **catalytic hydrogenation** turns nitriles into primary amines
- used to **ascend the amine series** — to make an amine with one carbon more than the starting amine



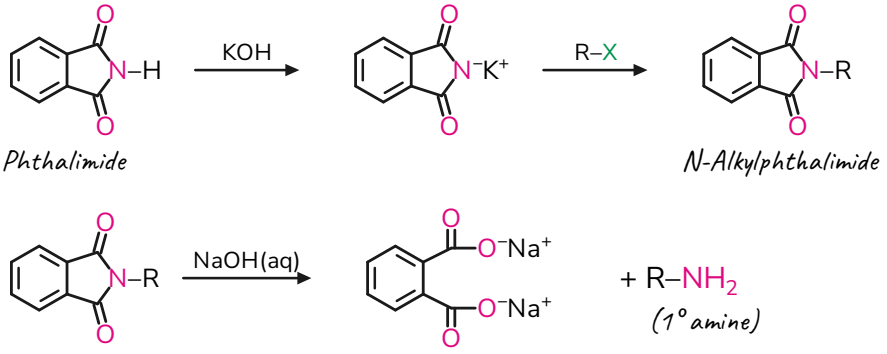
4 · Reducing amides

- amides reduced by **$LiAlH_4$** give amines



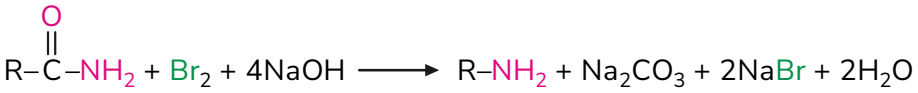
5 · Gabriel phthalimide synthesis

- a route to **primary amines**
- phthalimide + **ethanolic KOH** → potassium salt of phthalimide; heating this with an **alkyl halide**, then **alkaline hydrolysis**, releases the matching primary amine
- **aromatic primary amines cannot be made this way** — aryl halides do not undergo nucleophilic substitution with the phthalimide anion



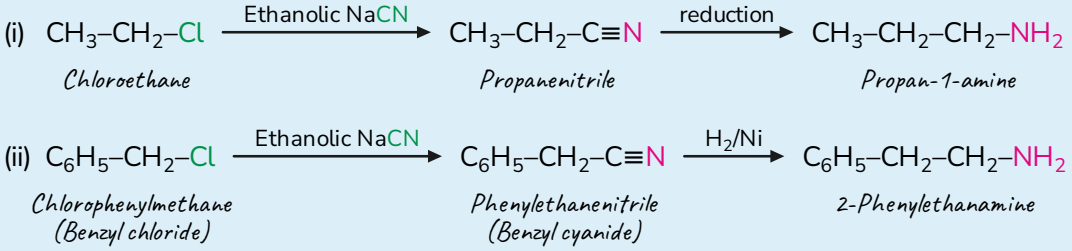
6 · Hoffmann bromamide degradation

- Hoffmann's method for primary amines: an **amide** is treated with **Br_2** in aqueous or ethanolic **NaOH**
- during this degradation an alkyl or aryl group **migrates** from the amide's carbonyl carbon to the nitrogen
- the amine formed has **one carbon fewer** than the amide it came from



Worked example 9.6

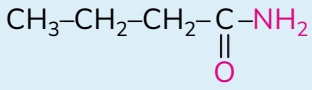
Show by equations how to convert (i) $CH_3–CH_2–Cl$ into $CH_3–CH_2–CH_2–NH_2$; (ii) $C_6H_5–CH_2–Cl$ into $C_6H_5–CH_2–CH_2–NH_2$.



Worked example 9.c

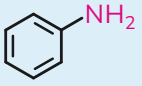
Draw and give IUPAC names for (i) the amide that yields propanamine in the Hoffmann bromamide reaction; (ii) the amine obtained when benzamide undergoes Hoffmann degradation.

- propanamine has **three** carbons, and the reaction removes one — so the amide must carry **four**



Butanamide

- benzamide is an aromatic amide with **seven** carbons; losing one leaves an aromatic primary amine with **six**



Aniline or benzenamine

In-text question

T3 How would you carry out these conversions?

(i) benzene to aniline
(ii) benzene to N,N-dimethylaniline
(iii) Cl–$(CH_2)_4$–Cl to hexane-1,6-diamine

A.5 Physical properties

State, odour, colour

- lower aliphatic amines are **gases** with a **fishy smell**
- primary amines of **three or more** carbons are liquids; still larger ones are solids
- aniline and other arylamines are normally colourless, but **darken on storage** through atmospheric oxidation

Solubility

- lower aliphatic amines dissolve in water because they **hydrogen-bond** with it
- as molar mass grows, the hydrophobic alkyl part grows too and solubility falls; higher amines are practically **insoluble**
- amines dissolve in organic solvents such as alcohol, ether and benzene

Amines against alcohols

- electronegativity: N of an amine **3.0**, O of an alcohol **3.5** — use this to predict how the two classes compare in water
- try it: which is more soluble in water, **butan-1-ol** or **butan-1-amine**, and why?
- alcohols are **more polar** than amines and form **stronger** intermolecular hydrogen bonds

Association and boiling point

- 1° and 2° amines **associate**: N of one molecule hydrogen-bonds to H of another
- association is greater in 1° amines — two H atoms on N are available — than in 2°
- 3° amines do not associate: no H on N is left for hydrogen bonding
- this sets the boiling-point order of isomeric amines, stated on sheet 8

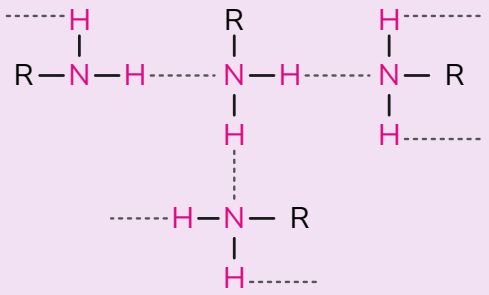


Figure 9.B Intermolecular hydrogen bonding in primary amines

Boiling points of isomeric amines

- order: **primary > secondary > tertiary**
- the N–H···N network of a primary amine is sketched in Figure 9.B
- Table 9.B sets amines beside an alcohol and an alkane of nearly the same molar mass

TABLE 9.B **Comparison of Boiling Points of Amines, Alcohols and Alkanes of Similar Molecular Masses**

No.	Compound	Molar mass	b.p./K
1	n-$C_4H_9NH_2$	73	350.8
2	$(C_2H_5)_2NH$	73	329.3
3	$C_2H_5N(CH_3)_2$	73	310.5
4	$C_2H_5CH(CH_3)_2$	72	300.8
5	n-C_4H_9OH	74	390.3

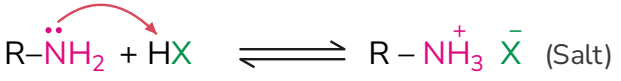
A.6 Chemical reactions

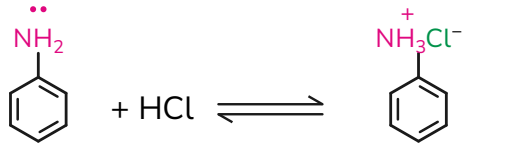
What makes amines reactive

- the **electronegativity difference** between N and H, together with the **unshared pair** on N
- the **number of H atoms on N** steers the reaction, so primary ($–NH_2$), secondary (>N–H) and tertiary (>N–) amines part ways in many reactions
- the unshared pair also lets amines act as **nucleophiles**

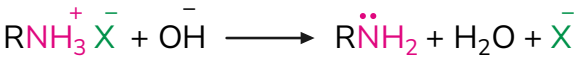
1 · Basic character

- amines are **basic**: with acids they form **salts**
- the N lone pair takes up the proton of the acid (curved arrow below)
- aniline and HCl give **anilinium chloride**





Aniline *Anilinium chloride*



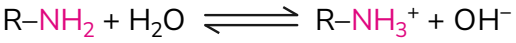
Getting the amine back

- treating an amine salt with a base such as **NaOH** regenerates the parent amine
- amine salts **dissolve in water** but **not** in organic solvents like ether
- this is the basis for **separating amines** from non-basic organic compounds that are insoluble in water

Why amines count as bases

- salt formation with mineral acids shows their basic nature
- the unshared pair on N makes them **Lewis bases**
- basic strength is best judged from K_b and pK_b

Base strength in water



$$K = \frac{[R\text{–}NH_3^+][OH^-]}{[R\text{–}NH_2][H_2O]}$$

$$\text{or } K[H_2O] = \frac{[R\text{–}NH_3^+][OH^-]}{[R\text{–}NH_2]}$$

$$\text{or } K_b = \frac{[R\text{–}NH_3^+][OH^-]}{[R\text{–}NH_2]} \qquad pK_b = -\log K_b$$

Reading K_b and pK_b

- the **larger K_b** — or the **smaller pK_b** — the stronger the base
- pK_b of ammonia: **4.75**
- **aliphatic amines** are stronger bases than NH_3: the **+I effect** of alkyl groups raises electron density on N; their pK_b values run from **3 to 4.22**
- **aromatic amines** are weaker bases than NH_3, because the aryl group withdraws electrons

TABLE 9.C pK_b Values of Amines in Aqueous Phase

Name of amine	pK_b
Methanamine	3.38
N-Methylmethanamine	3.27
N,*N*-Dimethylmethanamine	4.22
Ethanamine	3.29
N-Ethylethanamine	3.00
N,*N*-Diethylethanamine	3.25
Benzenamine	9.38
Phenylmethanamine	4.70
N-Methylaniline	9.30
N,*N*-Dimethylaniline	8.92

When inductive effect is not enough

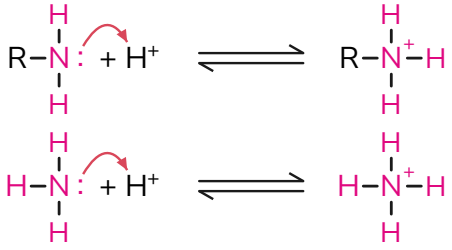
- judging K_b values from +I or –I effects alone throws up some **anomalies**
- other factors also shape basic strength — **solvation**, **steric hindrance** and more; they are unpacked below

Structure–basicity relationship

- basicity depends on how **readily the cation forms** when the amine takes a proton from an acid
- the **more stable the cation** relative to the amine, the **more basic** the amine

A.6 Chemical reactions · basicity continued

(a) Alkanamines against ammonia



- the alkyl group **releases electrons** towards N, so the unshared pair is more available to the acid's proton
- the substituted ammonium ion is further **stabilised** as the +I effect spreads out its positive charge
- so alkylamines are **stronger bases than NH_3**, and basicity ought to climb with each added alkyl group

Gas phase

- the expected trend holds: **3° amine > 2° amine > 1° amine > NH_3**

Aqueous phase

- the trend breaks down, as the pK_b values of Table 9.C show
- in water the cation is stabilised both by the **+I effect** and by **solvation** with water molecules
- the **larger the ion**, the less it is solvated and the less it is stabilised

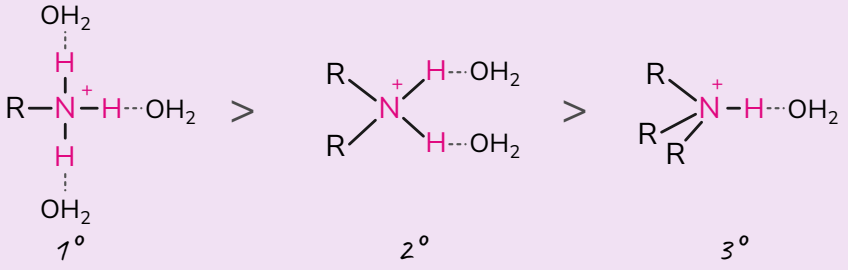


Figure 9.C Decreasing order of extent of H-bonding in water and order of stability of ions by solvation.

Solvation and steric effects

- a more stable substituted ammonium cation means a stronger parent base — so solvation alone would rank aliphatic amines **1° > 2° > 3°**, the reverse of the inductive order
- a small group such as $-CH_3$ puts **no steric hindrance** in the way of H-bonding; groups bigger than CH_3 do
- so switching $-CH_3$ for $-C_2H_5$ **changes the order** of basic strength

Net result in water

- a fine balance of **inductive effect**, **solvation** and **steric hindrance** sets the basic strength of alkyl amines in water
- ethyl series: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$
- methyl series: $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(b) Arylamines against ammonia

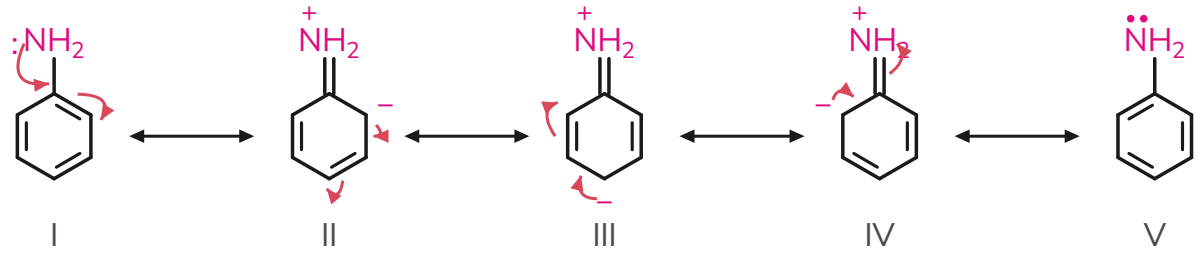
Why aniline has a high pK_b

- in arylamines the $-NH_2$ group sits directly on the benzene ring
- the unshared pair on N is therefore in **conjugation** with the ring and less available for protonation
- aniline is a **resonance hybrid of five structures** (I–V below)

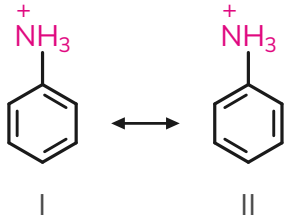
Substituted anilines

- **electron-releasing** groups such as $-OCH_3$ and $-CH_3$ **raise** basic strength
- **electron-withdrawing** groups such as $-NO_2$, $-SO_3H$, $-COOH$ and $-X$ **lower** it

Resonance in aniline



Anilinium ion



Aniline against the anilinium ion

- on taking a proton, the anilinium ion has **only two** resonance structures (Kekulé forms)
- more resonance structures → greater stability, so aniline (five) is more stable than the anilinium ion
- hence aniline and other aromatic amines accept protons less readily — they are **weaker bases than NH_3**

Worked example 9.d

Put these in decreasing order of basic strength: $C_6H_5NH_2$, $C_2H_5NH_2$, $(C_2H_5)_2NH$, NH_3.

Answer: $(C_2H_5)_2NH > C_2H_5NH_2 > NH_3 > C_6H_5NH_2$

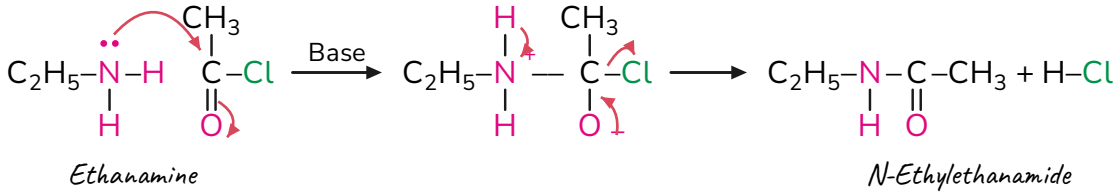
2 · Alkylation

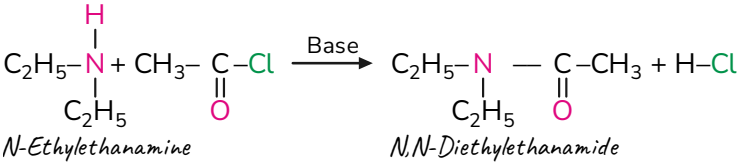
- amines are alkylated by alkyl halides (see Unit 6, Class XII)

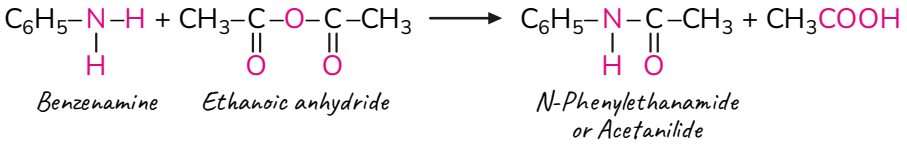
3 · Acylation

- aliphatic and aromatic **1° and 2°** amines react with **acid chlorides, anhydrides and esters** by nucleophilic substitution — this is **acylation**
- think of it as an acyl group taking the place of an H on $–NH_2$ or >N–H; the products are **amides**
- run in the presence of a base **stronger than the amine**, such as **pyridine**, which mops up the HCl and pushes the equilibrium to the right

Acylation, step by step

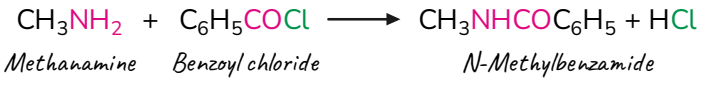






Benzoylation · acids

- reaction with **benzoyl chloride**, C_6H_5COCl, is called **benzoylation**
- carboxylic acids simply form **salts** with amines at room temperature



4 · Carbylamine reaction

- aliphatic *and* aromatic **primary** amines, heated with **chloroform** and **ethanolic KOH**, give **isocyanides** (carbylamines) — foul-smelling compounds
- secondary and tertiary amines give no such reaction
- known as the **carbylamine reaction** or **isocyanide test**; it is the test for primary amines

$$R\text{–}NH_2 + CHCl_3 + 3KOH \xrightarrow{\text{Heat}} R\text{–}NC + 3KCl + 3H_2O$$

5 · Reaction with nitrous acid

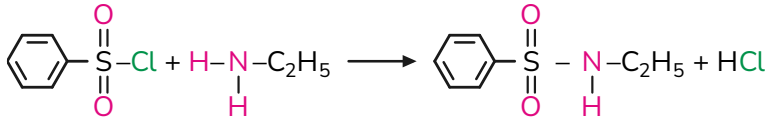
- nitrous acid is generated *in situ* from a mineral acid and sodium nitrite; the three classes of amine respond differently
- **(a) primary aliphatic** amines give aliphatic diazonium salts, which are unstable and break down to **alcohols** while releasing **N_2 quantitatively** — the measured N_2 is used to estimate amino acids and proteins
- **(b) aromatic** amines at **low temperature (273–278 K)** give **diazonium salts** — a key class for making many aromatic compounds (section B.1)
- secondary and tertiary amines react with nitrous acid in another way

$$R\text{–}NH_2 + HNO_2 \xrightarrow{NaNO_2 + HCl} [R\text{–}N_2^+Cl^-] \xrightarrow{H_2O} ROH + N_2 + HCl$$

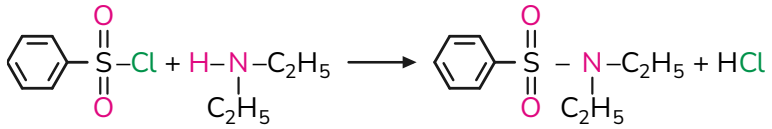
$$\underset{\text{Aniline}}{C_6H_5\text{–}NH_2} \xrightarrow[273\text{–}278\ K]{NaNO_2 + 2HCl} \underset{\text{Benzenediazonium chloride}}{C_6H_5\text{–}N_2^+Cl^-} + NaCl + 2H_2O$$

6 · Reaction with arylsulphonyl chloride

- benzenesulphonyl chloride, $C_6H_5SO_2Cl$ — **Hinsberg's reagent** — reacts with primary and secondary amines to give **sulphonamides**
- **(a) primary amine** (ethanamine here) → N-ethylbenzenesulphonamide; its N–H hydrogen is **strongly acidic** because the sulphonyl group pulls electrons hard, so the product **dissolves in alkali**
- **(b) secondary amine** (N-ethylethanamine here) → N,N-diethylbenzenesulphonamide



N-Ethylbenzenesulphonamide (soluble in alkali)



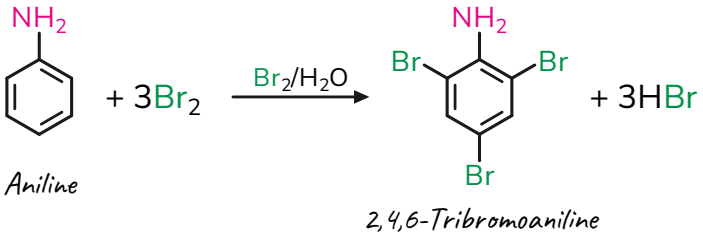
N,N-Diethylbenzenesulphonamide

(c) Tertiary amines · using the test

- N,N-diethylbenzenesulphonamide has **no H on N**, so it is not acidic and **does not dissolve in alkali**
- **tertiary** amines do **not react** with benzenesulphonyl chloride
- these three different responses let us **tell 1°, 2° and 3° amines apart** and **separate** a mixture of them
- today ***p*-toluenesulphonyl chloride** is used in place of benzenesulphonyl chloride

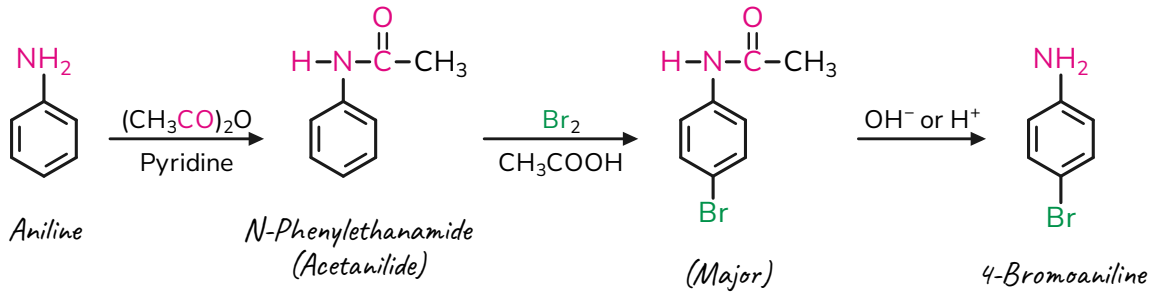
7 · Electrophilic substitution

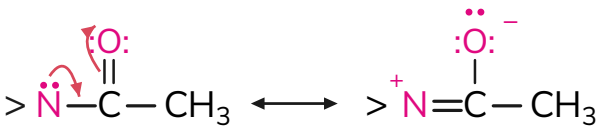
- of aniline's five resonance structures, the ***ortho* and *para*** positions carry the most electron density
- $-NH_2$ is therefore ***ortho*/*para*-directing** and a **powerful activating** group
- **(a) Bromination:** with bromine water at room temperature aniline gives a **white precipitate** of 2,4,6-tribromoaniline



Taming the $-NH_2$ group

- the difficulty: aromatic amines are **very reactive**, and substitution heads for every *ortho* and *para* site
- to get a **monosubstituted** product, **protect** $-NH_2$ by **acetylation** with acetic anhydride, substitute, then **hydrolyse** the amide back to the amine
- in acetanilide the N lone pair is drawn towards the carbonyl oxygen by resonance (below)



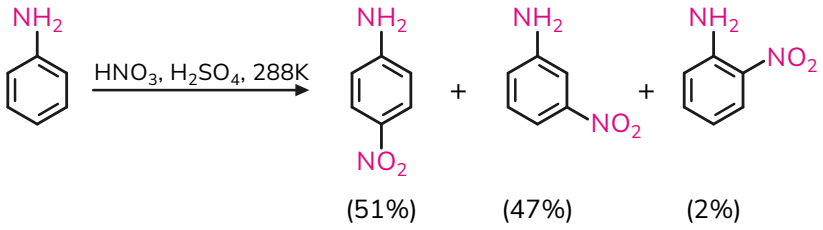


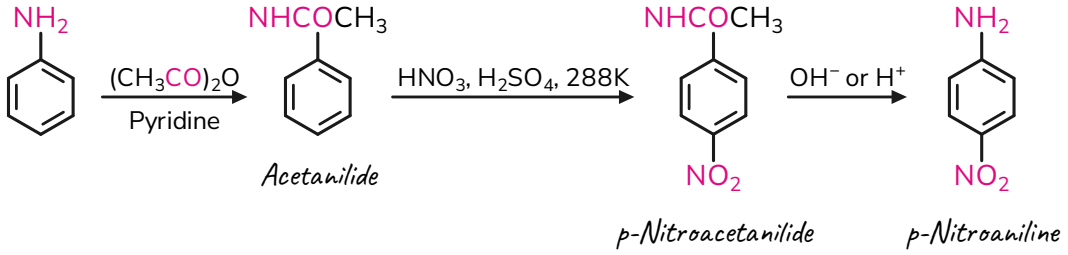
Acetanilide is less activating

- the N lone pair is less free to feed electrons into the benzene ring by resonance
- so –$NHCOCH_3$ **activates the ring less** than –NH_2 does

(b) Nitration

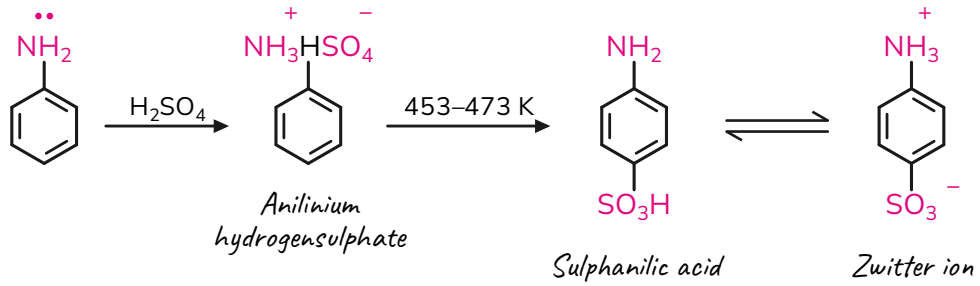
- direct nitration gives **tarry oxidation products** alongside the nitro compounds
- in the strongly acidic medium aniline is **protonated** to the anilinium ion, which is ***meta*-directing** — so a **large share of the *meta* product** forms beside *ortho* and *para*
- protecting –NH_2 first by acetylation with acetic anhydride keeps the nitration under control and makes the ***p*-nitro** compound the main product





(c) Sulphonation

- concentrated H_2SO_4 converts aniline to **anilinium hydrogensulphate**
- heating this with H_2SO_4 at **453–473 K** gives mainly *p*-aminobenzenesulphonic acid — **sulphanilic acid**



No Friedel–Crafts reaction

- aniline gives **no** Friedel–Crafts alkylation or acetylation
- it forms a **salt with $AlCl_3$**, the Lewis-acid catalyst; the N becomes positively charged and turns into a **strongly deactivating** group

In-text questions

T4 Place in increasing order of basic strength:

(i) $C_2H_5NH_2$, $C_6H_5NH_2$, NH_3, $C_6H_5CH_2NH_2$ and $(C_2H_5)_2NH$

(ii) $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$, $C_6H_5NH_2$

(iii) CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$, $C_6H_5NH_2$, $C_6H_5CH_2NH_2$

T5 Finish these acid–base reactions and name what forms: (i) $CH_3CH_2CH_2NH_2 + HCl \rightarrow$ (ii) $(C_2H_5)_3N + HCl \rightarrow$

T6 Aniline is alkylated with excess methyl iodide in sodium carbonate solution. Write the reactions leading to the final product.

T7 Give the equation for aniline reacting with benzoyl chloride, and name the product.

T8 Draw the isomers with formula C_3H_9N. Give IUPAC names for those that release nitrogen gas with nitrous acid.

II · Diazonium salts

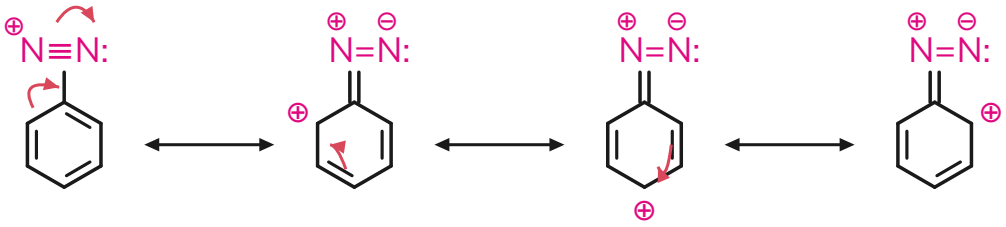
Formula and naming

- general formula $RN_2^+X^-$: R is an **aryl** group; X^- can be Cl^-, Br^-, HSO_4^-, BF_4^- and so on
- name = parent hydrocarbon + **diazonium** + name of the anion (chloride, hydrogensulphate ...); N_2^+ is the **diazonium group**
- $C_6H_5N_2^+Cl^-$ benzenediazonium chloride; $C_6H_5N_2^+HSO_4^-$ benzenediazonium hydrogensulphate

How stable are they?

- **primary aliphatic** amines give **highly unstable** alkyldiazonium salts (see A.6)
- **primary aromatic** amines give arenediazonium salts that last a **short time in solution at 273–278 K**
- the arenediazonium ion owes this stability to **resonance** (below)

Resonance in the arenediazonium ion



B.1 Preparing diazonium salts

- benzenediazonium chloride is made from **aniline** and **nitrous acid at 273–278 K**; the nitrous acid is produced in the flask from **$NaNO_2$ and HCl**
- turning a primary aromatic amine into its diazonium salt is called **diazotisation**
- being unstable, the salt is usually **not stored** but used straight after it is made

$$C_6H_5NH_2 + NaNO_2 + 2HCl \xrightarrow{273–278\ K} C_6H_5N_2^+Cl^- + NaCl + 2H_2O$$

B.2 Physical properties

- **benzenediazonium chloride**: colourless crystalline solid, **very soluble in water**; stable in the cold but **reacts with water when warmed**; **decomposes readily when dry**
- **benzenediazonium fluoroborate**: **insoluble in water** and **stable at room temperature**

B.3 Chemical reactions

Two families of reactions

- **(A)** reactions in which **nitrogen is displaced**
- **(B)** reactions in which the **diazo group is kept**

A · Displacement of nitrogen

- the diazonium group is an **excellent leaving group**; Cl^-, Br^-, I^-, CN^- and OH^- push out the nitrogen from the ring
- the N_2 formed **escapes as a gas**

1 · Halide or cyanide

- Cl^-, Br^- and CN^- enter the ring easily in the presence of **Cu(I)** — the **Sandmeyer reaction**
- alternatively, Cl or Br can be introduced with the matching **halogen acid and copper powder** — the **Gattermann reaction**
- Sandmeyer gives the **better yield**

Sandmeyer

$$ArN_2^+X^- \xrightarrow{Cu_2Cl_2/HCl} ArCl + N_2$$
$$ArN_2^+X^- \xrightarrow{Cu_2Br_2/HBr} ArBr + N_2$$
$$ArN_2^+X^- \xrightarrow{CuCN/KCN} ArCN + N_2$$

Gattermann

$$ArN_2^+X^- \xrightarrow{Cu/HCl} ArCl + N_2 + CuX$$
$$ArN_2^+X^- \xrightarrow{Cu/HBr} ArBr + N_2 + CuX$$

2 · Iodide

- iodine does not go into the benzene ring easily by direct reaction
- treating the diazonium salt solution with **KI** gives **iodobenzene**

3 · Fluoride

- arenediazonium chloride + **fluoroboric acid** precipitates **arenediazonium fluoroborate**
- heating it breaks it down to the **aryl fluoride**

$$ArN_2^+Cl^- + KI \longrightarrow ArI + KCl + N_2$$

$$ArN_2^+Cl^- + HBF_4 \longrightarrow Ar{-}N_2^+BF_4^- \xrightarrow{\Delta} Ar{-}F + BF_3 + N_2$$

4 · Hydrogen

- mild reducing agents — **hypophosphorous acid** (phosphinic acid) or **ethanol** — reduce diazonium salts to **arenes**
- they are themselves oxidised, to **phosphorous acid** and **ethanal** respectively (equations on sheet 18)

$$ArN_2^+Cl^- + H_3PO_2 + H_2O \longrightarrow ArH + N_2 + H_3PO_3 + HCl$$

$$ArN_2^+Cl^- + CH_3CH_2OH \longrightarrow ArH + N_2 + CH_3CHO + HCl$$

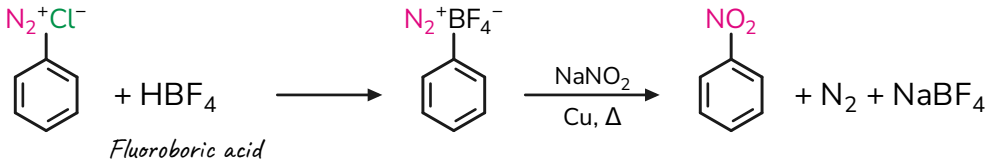
5 · Hydroxyl group

- if the diazonium salt solution warms to **283 K**, the salt is **hydrolysed to phenol**

$$ArN_2^+Cl^- + H_2O \longrightarrow ArOH + N_2 + HCl$$

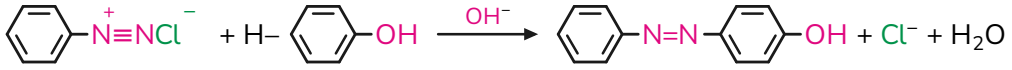
6 · Nitro group

- heating the **diazonium fluoroborate** with aqueous **$NaNO_2$** in the presence of **copper** replaces the diazonium group by $-NO_2$

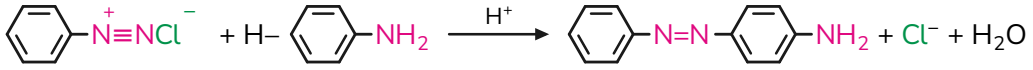


B · Diazo group kept: coupling reactions

- the **azo** products have an **extended conjugated system** — both aromatic rings are linked through –N=N–
- such compounds are usually **coloured** and serve as **dyes**
- benzenediazonium chloride couples with **phenol** at phenol's *para* position → ***p*-hydroxyazobenzene**; with **aniline** it gives ***p*-aminoazobenzene** — this kind of reaction is **coupling**
- coupling is an **electrophilic substitution** reaction



p-Hydroxyazobenzene (orange dye)



p-Aminoazobenzene (yellow dye)

B.4 Diazonium salts in aromatic synthesis

What diazonium salts make possible

- handy intermediates for putting **–F, –Cl, –Br, –I, –CN, –OH and $-NO_2$** onto an aromatic ring
- aryl **fluorides and iodides** cannot be made by direct halogenation

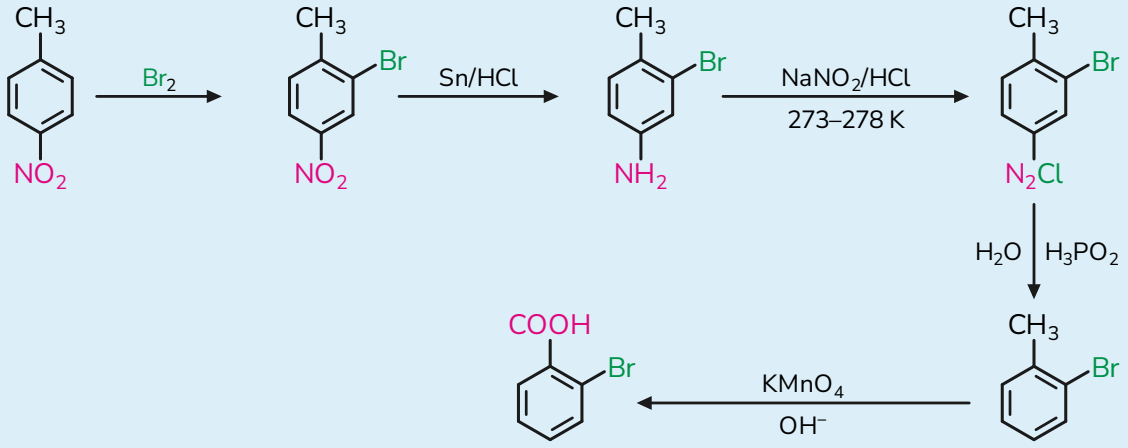
Cyano group

- –CN cannot be introduced by nucleophilic substitution of Cl in chlorobenzene
- cyanobenzene is, however, easily made from a diazonium salt

...so replacing the diazo group is the way to make substituted aromatic compounds that **cannot be obtained by direct substitution** in benzene or a substituted benzene.

Worked example 9.e

Outline how 4-nitrotoluene can be converted into 2-bromobenzoic acid.



In-text question

T9 Carry out these conversions: (i) 3-methylaniline to 3-nitrotoluene; (ii) aniline to 1,3,5-tribromobenzene.

Summary · what amines are

- amines are ammonia derivatives in which alkyl or aryl groups stand in for H atoms
- one H replaced → **primary**, R-NH_2; **secondary** R_2NH or R-NHR'; **tertiary** R_3N, RNR'R" or R_2NR'
- 2° and 3° amines are **simple** when the groups are identical and **mixed** when they differ
- all three classes, like NH_3, carry one unshared pair on N and so act as **Lewis bases**

Summary · making them, basicity

- usually prepared from **nitro compounds, halides, amides, imides** and the like
- **hydrogen bonding** governs their physical properties
- in **alkylamines**, electron release, steric factors and H-bonding together decide how stable the substituted ammonium cation is in protic polar solvents — and so the basicity; alkyl amines are **stronger bases than NH_3**
- in **aromatic amines**, electron-releasing groups raise and electron-withdrawing groups lower basicity; **aniline is a weaker base** ... (continued on sheet 20)

Summary · reactions of amines

- … than ammonia. Amine reactions are controlled by how available the **unshared pair on N** is
- the **number of H atoms on N** decides the type of reaction and product — the basis for identifying and telling apart 1°, 2° and 3° amines; ***p*-toluenesulphonyl chloride** is used for this identification
- an amino group **raises the reactivity** of an aromatic ring; **acylation** with acetyl chloride or acetic anhydride keeps it in check
- tertiary amines such as **trimethylamine** serve as **insect attractants**

Summary · aryldiazonium salts

- normally prepared from **arylamines**
- the diazonium group is replaced by many nucleophiles — a convenient route to **aryl halides, cyanides, phenols**, and to **arenes** by reductive removal of the diazo group
- coupling with **phenols or arylamines** gives **azo dyes**

Exercises

X1 Give the IUPAC name of each compound and label it primary, secondary or tertiary.

(i) $(CH_3)_2CHNH_2$ (ii) $CH_3(CH_2)_2NH_2$ (iii) $CH_3NHCH(CH_3)_2$

(iv) $(CH_3)_3CNH_2$ (v) $C_6H_5NHCH_3$ (vi) $(CH_3CH_2)_2NCH_3$

(vii) *m*–$BrC_6H_4NH_2$

X2 Suggest one chemical test that tells apart the two members of each pair.

(i) methylamine and dimethylamine (ii) secondary and tertiary amines

(iii) ethylamine and aniline (iv) aniline and benzylamine

(v) aniline and N-methylaniline

X3 Explain each observation.

(i) aniline has a higher pK_b than methylamine

(ii) ethylamine dissolves in water, but aniline does not

(iii) methylamine in water precipitates hydrated ferric oxide from ferric chloride

(iv) the amino group directs *o*/*p* in aromatic electrophilic substitution, yet nitration of aniline gives a large amount of *m*-nitroaniline

(v) aniline undergoes no Friedel–Crafts reaction

(vi) diazonium salts of aromatic amines are more stable than those of aliphatic amines

(vii) Gabriel phthalimide synthesis is the preferred way to make primary amines

X4 Put in order:

(i) decreasing pK_b: $C_2H_5NH_2$, $C_6H_5NHCH_3$, $(C_2H_5)_2NH$ and $C_6H_5NH_2$

(ii) increasing basic strength: $C_6H_5NH_2$, $C_6H_5N(CH_3)_2$, $(C_2H_5)_2NH$ and CH_3NH_2

(iii) increasing basic strength: (a) aniline, *p*-nitroaniline and *p*-toluidine; (b) continued on sheet 21

X4 continued

(iii) (b) $C_6H_5NH_2$, $C_6H_5NHCH_3$, $C_6H_5CH_2NH_2$

(iv) decreasing basic strength in the gas phase: $C_2H_5NH_2$, $(C_2H_5)_2NH$, $(C_2H_5)_3N$ and NH_3

(v) increasing boiling point: C_2H_5OH, $(CH_3)_2NH$, $C_2H_5NH_2$

(vi) increasing solubility in water: $C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$

X5 Show how each conversion can be made.

(i) ethanoic acid → methanamine

(ii) hexanenitrile → 1-aminopentane

(iii) methanol → ethanoic acid

(iv) ethanamine → methanamine

(v) ethanoic acid → propanoic acid

(vi) methanamine → ethanamine

(vii) nitromethane → dimethylamine

(viii) propanoic acid → ethanoic acid

X6 Give a method for identifying primary, secondary and tertiary amines, with the equations involved.

X7 Write brief notes on:

(i) carbylamine reaction

(ii) diazotisation

(iii) Hofmann's bromamide reaction

(iv) coupling reaction

(v) ammonolysis

(vi) acetylation

(vii) Gabriel phthalimide synthesis

X8 Carry out these conversions:

(i) nitrobenzene → benzoic acid

(ii) benzene → *m*-bromophenol

(iii) benzoic acid → aniline

(iv) aniline → 2,4,6-tribromofluorobenzene

(v) benzyl chloride → 2-phenylethanamine

(vi) chlorobenzene → *p*-chloroaniline

(vii) aniline → *p*-bromoaniline

(viii) benzamide → toluene

(ix) aniline → benzyl alcohol

X9 Identify A, B and C in each sequence.

(i) $CH_3CH_2I \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow{NaOH + Br_2} C$

(ii) $C_6H_5N_2Cl \xrightarrow{CuCN} A \xrightarrow{H_2O/H^+} B \xrightarrow[\Delta]{NH_3} C$

(iii) $CH_3CH_2Br \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[0°C]{HNO_2} C$

(iv) $C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273\ K]{NaNO_2 + HCl} B \xrightarrow[\Delta]{H_2O/H^+} C$

(v) $CH_3COOH \xrightarrow[\Delta]{NH_3} A \xrightarrow{NaOBr} B \xrightarrow{NaNO_2/HCl} C$

(vi) $C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273\ K]{HNO_2} B \xrightarrow{C_6H_5OH} C$

Exercises continued

X10 An aromatic compound A, heated with aqueous ammonia, gives B. Heating B with Br_2 and KOH gives C, of formula C_6H_7N. Give the structures and IUPAC names of A, B and C.

X11 Complete these reactions:

(i) $C_6H_5NH_2 + CHCl_3 + alc.KOH \rightarrow$

(ii) $C_6H_5N_2Cl + H_3PO_2 + H_2O \rightarrow$

(iii) $C_6H_5NH_2 + H_2SO_4$ (conc.) $\rightarrow$

(iv) $C_6H_5N_2Cl + C_2H_5OH \rightarrow$

(v) $C_6H_5NH_2 + Br_2(aq) \rightarrow$

(vi) $C_6H_5NH_2 + (CH_3CO)_2O \rightarrow$

(vii) $C_6H_5N_2Cl$ treated with (i) HBF_4, then (ii) $NaNO_2/Cu$, $\Delta \rightarrow$

X12 Why can aromatic primary amines not be made by Gabriel phthalimide synthesis?

X13 Give the reactions of (i) aromatic and (ii) aliphatic primary amines with nitrous acid.

X14 Suggest a reason for each:

(i) amines are less acidic than alcohols of similar molecular mass

(ii) primary amines boil at a higher temperature than tertiary amines

(iii) aliphatic amines are stronger bases than aromatic amines

Answers to some in-text questions · T4

- (i) $C_6H_5NH_2 < NH_3 < C_6H_5CH_2NH_2 < C_2H_5NH_2 < (C_2H_5)_2NH$
- (ii) $C_6H_5NH_2 < C_2H_5NH_2 < (C_2H_5)_3N < (C_2H_5)_2NH$
- (iii) $C_6H_5NH_2 < C_6H_5CH_2NH_2 < (CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$